MedAgent Clinical Decision Support

Case ID: edit_gate_case | **Status:** EDIT | **Physician:** Dr. Specialist | **Date:** 2026-10-16 06:53 UTC

CRITICAL DISCLAIMER: Decision-support draft generated by AI agents. It is not a diagnosis and must be reviewed, corrected and approved by a licensed physician before any clinical use.

Patient Intake & Presentation

Patient: Age: Unspecified, Sex: Unspecified
Notes & History: Fever and productive cough.

Clinical Summary / Impression

Physician edited summary: Community acquired bacterial pneumonia confirmed.

Ranked Differential Diagnoses

#	Condition	Likelihood	Key Supporting Findings	Citations
1	**Community-Acquired Pneumonia**	HIGH	fever, cough, Leukocytosis, Leukocytosis	S1, S3
2	**Acute Viral Bronchitis / Atypical Infection**	LOW	Respiratory presentation, Cough and dyspnea	S2

Laboratory Findings & Identified Patterns

Analyte / Test	Result	Unit	Reference Interval	Flag
WBC Count	**14.5**	x10^3/uL	4.5 – 11	HIGH

Inferred Diagnoses (ClinicalLabNet Neural Classifier):
• **[MODERATE] Type 2 Diabetes Mellitus / Severe Hyperglycemia** [60.0% Neural Confidence] (Likelihood: LOW): Neural Network Confidence: 60.0% (ClinicalLabNet deep multi-label classifier). Biomarker pattern is characteristic of type 2 diabetes mellitus / severe hyperglycemia. *Pathophysiology:* Chronic metabolic disorder of peripheral insulin resistance and progressive pancreatic beta-cell secretory failure resulting in sustained hyperglycemia.

Synthesized Clinical Patterns:
• **[ROUTINE] Leukocytosis:** Elevated white blood cell count indicating active inflammatory, infectious, or reactive marrow response.

Suggested Diagnostic Workup

• Sputum gram stain and culture
• Blood cultures prior to antibiotics
• Follow-up chest radiograph in 6 weeks
• Respiratory viral multiplex PCR

Physician Attestation: I, Dr. Specialist, hereby confirm that I have reviewed, verified, and adjusted this decision-support synthesis. Decision: **EDIT**.
Reviewer Notes: Adjusted summary for bacterial etiology.
Signature: _____ **Date:** 2026-10-16 06:53 UTC